

7th International Conference on Intelligent, Interactive Systems and Applications

Research and Application of Network Anomaly Traffic Detection System

Xin Yue^a, Guangming Bo^{b,*}, Jianxun Zhang^a

^a*School of information technology and engineering, Tianjin University of Technology and Education, Tianjin, 300222, China*

^b*Division of state-owned assets management, Tianjin University of Technology and Education, Tianjin, 300222, China*

Abstract

This paper puts forward a network security computing platform which based on the open source big data technology. This network anomaly traffic detection system includes a comprehensive and effective traffic anomaly detection algorithm. The algorithm combines the exponentially weighted moving average algorithm (EWMA) and the anomaly flow interval mapping algorithm two methods. The test experiments contain the physical cluster building parameters, the performance of the system and the DDoS attack verification. It is verified by the Tianjin Education metropolitan area network traffic and the NSFOCUS against denial attack event log service system (NSFOCUS ANTI-DDoS system). In Particular, the network anomaly traffic detection system can accurately identify the occurrence period of anomaly traffic such as the flow direction and the event priority. What's more, it has strong fault tolerance which can achieve error recovery in a short time. All the test experiment results can verify that the system can effectively detect anomaly traffic and real-time monitor the dynamic network.

© 2022 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<https://creativecommons.org/licenses/by-nc-nd/4.0>)

“Peer-review under responsibility of the scientific committee of the 7th International Conference on Intelligent, Interactive Systems and Applications”

Keywords: Anomaly detection; DDoS attack; network traffic; Real-time monitor

1. Introduction

With the rapid development of computer technology and the continuous deepening of network applications, computer network plays an important role in all the persons daily life. At the same time, the network is also facing a large number of threats [1]. Large scale worm outbreak, distributed denial of service and the network service disruption had threaten the normal Internet operation [2]. In the event of scale network security, the network traffic

* Corresponding author. Tel.: 022-88286089; fax: 022-88286089.

E-mail address: yuxin92@163.com(X. Yue), gzclab@tute.edu.cn(G. Bo), zhangjx@tute.edu.cn(J. Zhang)

will produce the corresponding changes. If you can quickly check the anomaly change, the network manager can improve the exception ability. Anomaly detection system is seriously be attended by researchers and industry professionals to detect unknown malicious attacks.

Wagner who come from the Swiss Federal Institute of Arno developed a backbone network traffic monitoring and analysis framework that called UPFrame [3]. Vietnam Quang Anh Tran proposed an improved neural network algorithm based on (BBNN, Block-Based Neural Network) the real-time intrusion detection system to analysis data flow characteristics [4]. Gu Wei designed a distributed data stream real-time computing platform Stream system which can deal with a variety of complex computing needs [5]. Gang Zhang realized a causal association analysis by the MapReduce programming model. It alerted the redundant modules such as network intrusion scene reconstruction system [6]. Hu Yan realized a large scale network detection system based on a Hierarchical Intrusion Cooperation Model [7]. Lihua Miao achieved a system which can detect the whole network real-time SYN flood attack. On the basis of a predefined DDoS attack scenario model, the system uses the algorithm called WSAND to classify real-time Network traffic data [8].

With the whole system architecture, this paper puts forward a network security computing platform which based on the open source big data technology. With the help of a variety of middleware technology, this platform also have real-time computing, massive data off-line processing map ability and the distributed machine learning ability. It can meet a variety of network security analysis scenarios of data collection and storage needs. In order to validate the feasibility of the system scheme, this paper on the platform achieve a real-time traffic monitoring system for Tianjin Education metropolitan area network. The traffic monitoring system can acquire online real-time load flow index and anomaly fluctuations, what's more, distinguish various DDoS attack detection and recognition. In conclusion, the main innovations of this paper is included as following.

1. Kafka and HBase is the primary option in our platform. What's more, the system has a outstanding scalability. Only increasing the physical machine number, the input data can be done by the upper layer applications.
2. The big data computing system with network security has been illustrated. The traffic monitoring system can realize the detection and identify various DDoS attacks. The system realized by the real-time network index, traffic anomaly fluctuation, and the attack types discrimination.
3. On the basis of large data computing platform, the DDoS detection algorithm is applied in the processing and analysed by Tianjin Educational Metropolitan area real-time network traffic. The performances of DDoS attacks detection have great effectiveness in depth analysis and discussion. All the test results show that the system has faster processing speed with higher detection accuracy.

2. Related technology theory

2.1. Exponentially weighted moving average algorithm

Exponentially Weighted Moving Average algorithm (EWMA) [9] is a commonly used sequence data processing algorithm. The method decreases to numerical weights with exponentially. The rules in time data are higher weights near the current point while smaller weights away from the current data. The core idea of EWMA algorithm is to calculate current value estimates before a moment estimation value and the current moment observables [10-12]. The following formula can be obtained by applying EWMA algorithm:

$$\bar{x}_t = \alpha \times x_t + (1 - \alpha) \times \bar{x}_{t-1} \quad (1)$$

$$e_t = \bar{x}_t - x_t \quad (2)$$

$\bar{x}$ respects the t moment weighted average, x_t represents the t time measured value. At the same time, α less than 1. In order to calculate the t time weighted average value, the past 2 hours data are brought into account to statistics. The weight factor parameter is set to $N = 120$. Formula 2 is used to calculate the t moment estimated error. After achieving all the parameters, we can calculate the real-time loading index upper bound at the current time.

2.2. Implementation of traffic anomaly location algorithm

Whenever a new time window arrival (60 seconds for a time window), the algorithm program on the spark calculates the time limitation and determine load index changes changeDirection(the direction of change). The role of the network is to determine whether the emergence of a burst of traffic jam grow up. Changing direction in the algorithm is mainly used to achieve the change in the flow rate of the anomaly location [13]. A successful implementation of the attack traffic will usually make the network load index experienced a time from a substantial growth to decline during the evolution [14]. The main strategies adopted by this algorithm combine EWMA with achieve fast positioning of anomaly intervals. By comparing the current load index and a load time window index, key indicators record_creation_num(the creation number) is experiencing from low to high jump. The actual measured value is greater than bygone. The program will execute the algorithm 2-1. The calculation logic as shows in Table 1:

Table 1. Anomaly traffic interval location algorithm.

Algorithm 2-1: anomaly traffic interval location algorithm

```

IF latestStartPoint == Null || latestStartPoint.priority < newItem.priority
IF currentAbnormalId != Null && latestStartPoint != Null
    storageManager.unmarkRawRecordById(currentAbnormalId)
    storageManager.unmarkRealtimeMetricsById(currentAbnormalId)
END IF

Initialize a new SuspiciousAbnormalIntervalPoint Instance
SuspiciousAbnormalIntervalPoint.timeFrame= RealtimeMetrics.timeFrame
SuspiciousAbnormalIntervalPoint.edgeType = START_POINT
SuspiciousAbnormalIntervalPoint.metrics = RealtimeMetrics
SuspiciousAbnormalIntervalPoint.judgement = Judgement

latestStartPoint = SuspiciousAbnormalIntervalPoint
currentAbnormalId = ABNORMAL_RECORD_ID_PREFIX + "yyyyMMddHHmm"

storageManager.unmarkRawRecord(RealtimeMetrics.timeFrame, currentAbnormalId)
storageManager.unmarkRealtimeMetrics(RealtimeMetrics.timeFrame , currentAbnormalId)

```

3. Experiment

This section introduces the specific experiment process. It contains the physical cluster building parameters, the performance of the system, and the DDoS attack verification.

3.1. Physical cluster building

The platform is been tested by four computer distributed clusters. It is said that the Hadoop distributed file system (HDFS) [15-16] is composed by four nodes. The host 311b-node-2 also play two roles that both namenode and datanode. The remaining three sets of host only run datanode background processes. 4 datanode nodes provide a total of 5.73TB disk storage space. In the massive structured data storage scenarios, this article uses the HBase1.1.2 column storage database. In order to improve the usability of the system, according to the official document of the proposal, this paper in the parameters "hbase.rootdir" specified HDFS as the underlying storage service providers, so that you can graft HBase in HDFS. Data integrity and availability was not host failure. The hardware configuration of the machine is shown in Table 2.

Table 2. Experimental environment machine configuration.

Host name	CPU	Memory	Hard disk drive	Network card
311b-node-2	Intel Xeon E5-2620 2.4GHz	64GB DDR3	3.4TB SATA	Broadcom BCM5720 Gigabit PCIe
B311	Intel Xeon X5670 2.93GHz	8GB DDR2	524GB SATA	Intel 82574L Gigabit Network Connection
310b-node-1	Intel Xeon E5-3603 1.8GHz	16GB DDR2	748GB SATA	Broadcom BCM5720 Gigabit PCIe
310b-node-2	Intel Xeon	16GB DDR2	645GB SATA	Intel 82579LM Gigabit Network Connection

3.2. Validation of the storm cluster performance

In the platform architecture designed in this paper, Storm computing engine undertakes a number of important real-time computing tasks. It includes data filtering, the conversion of flow data to structured data, real-time event monitoring, blacklist interception, real-time incremental statistics, etc. On the occasion of high concurrency with large data volume, Storm processing performance will have a very important impact on the overall speed of platform applications. If the Storm computing speed cannot keep up with the speed of data inflow, the data will accumulate. With the increase of processing delay, downstream business logic will inevitably be affected and the system availability will be greatly reduced. The throughput of storm distributed cluster completely stored the netflow data from 00:00:00 on December 13,2015 to 23:59:59 on December 13,2015. The data set was used as input to stress test the netflow real-time processing module introduced in all this paper. The test data used in this paper contains 52835120 netflow records, covering multiple c-class network segments in the Tianjin Education metropolitan area.

This article starts four worker nodes in the Storm cluster, while one worker instance running on each computer. The executor concurrent numbers for the spout and bolt are set as shown in Figure 1. The 24 hours of network traffic data took 29 minutes and 55 seconds (from 7:39:30, December 16,2015 to 8:09:25, December 16,2015). Fig.1. depicts the system throughput curve in minutes. Regardless of the initialization phase (7:39:00) and the end phase (8:09:00), the average throughput of the real-time traffic monitoring system is 1650067 netflow records per minute, with the fastest processing speed (1782258 records) and the slowest (1405952 records) occurring in the ninth and 21st time windows, respectively. Viewing the system log authors of the Storm records found that the netty client process responsible for transferring data on the host 310b-node-1 was suspended at around 7:57. It causes a partial loss of data sent by the upstream kafka-spout to the host worker instance. Thus it affects the system throughput. The supervisor, which monitors the running status of the host, tries to restart the netty client immediately after the event. The kafka-spout uses the ack timeout monitoring mechanism to detect data replay, so that the system eventually returns to normal with the collaboration of supervisor and spout. It can be seen from the figure that Storm has strong fault tolerance, which can achieve error recovery in a short time.

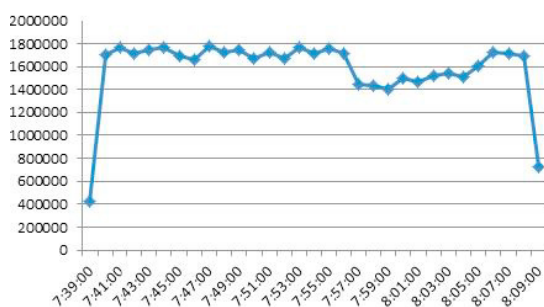


Fig. 1. Minutes-level system throughput.

For Storm, the worker concurrency number is another important factor affecting the system performance. In this paper, a set of test scenarios with different worker nodes to investigate the relationship between the number of

worker and system performance. By modifying the Storm cluster configuration, this article starts two worker instances on each computer, so that the system can take up to eight workers during operation (1024mb of available memory per worker). In this paper, a data set containing about 1.3 million netflow records is used as input to test the system when the worker number is 1,2,4, and 8, respectively. The change of the system running time is shown in Fig. 2. If only 1 worker is enabled, the network traffic real-time processing system takes 10 minutes and 55 seconds to complete the calculation of all the input data. When the number of concurrent working nodes is increased to 2, the system processing time is shortened to 7 minutes and 14 seconds, and the processing speed increases by 33.7%. When the number of worker s is set to 8, the system only takes 3 minutes and 21 seconds to generate the computing results and write it into the database. Compared with the single worker environment, the computing power of the system is increased by 69.3%. Therefore, according to the test results of this paper, the system calculation speed increases on average by 31.2% whenever the worker concurrency number is doubled.

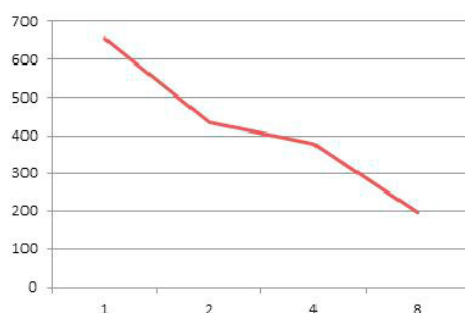


Fig. 2. effect of the worker concurrency number on the system operation time.

3.3. DDoS attack detection module verification

This paper evaluates the detection capability of the system based on the NSFOCUS anti denial of service System (NSFOCUS Anti-DDoS). The attack events recorded by ADS NSFOCUS mainly include the time stamp, the attack type, the source IP, the destination IP and other information.

DDoS detection system processes the Netflow TERNET data and the system's detection results. The validation data sets were compared and analyzed. The detection accuracy is shown in Table 3.

Table 3 Statistics of DDoS attack.

Data	Connection-Flood	ICMP-Flood	Stream-Flood	SYN-Flood	UDP-Flood
12.24	100%	92.9%	66.7%	54.5%	100%
12.25	100%	73.1%	66.7%	53.8%	100%
12.26	100%	77.8%	100%	72.2%	75%
12.27	100%	100%	100%	56.3%	100%
12.28	100%	80%	100%	56.3%	100%
12.29	100%	66.7%	100%	71.4%	100%

Connection-flood attacks and UDP-flood attacks against special ports are the most easily identified and discovered in the above five kinds of DDoS. The system has a very high detection success rate for these two DDoS attacks. In the analysis of the results of the detection of Stream-Flood, author found ADS system exists some false positives, through contrast ACK-flood attack detection results and the original Netflow data, the author thinks actually some Stream-Flood "attacker" in ADS log at the same period of ACK-Flood attacks, and ACK-Flood attack initiator is the false alarm in the event of the "victims".

In this paper, the average detection rate of ICMP-Flood attacks and SYN-flood attacks are 81.7% and 60.7% respectively. After in-depth analysis of the results of SYN-Flood detection, the authors found a new type of SYN attack, which is different from common attack methods. In this type of attack, the attacker sends a few flow SYN

(usually less than 3) per minute to the victim, but the entire attack will last for several hours. On the December 24th 2015, it observed by the author to the time duration for 9 hours 27 minutes of SYN Flood attack (as seen in Fig. 3.)

```
[date=2015-12-24 07:18:42, srcIp=212.129.27, srcPort=43324, dstIp=59.67.152, dstPort=46992, attackInfo=SYN-Flood, flag=SYN_INVALID]
[date=2015-12-24 07:18:42, srcIp=212.129.27, srcPort=43324, dstIp=59.67.152, dstPort=46992, attackInfo=SYN-Flood, flag=SYN_INVALID]
[date=2015-12-24 07:20:13, srcIp=212.129.27, srcPort=43324, dstIp=59.67.152, dstPort=47137, attackInfo=SYN-Flood, flag=SYN_INVALID]
[date=2015-12-24 07:20:44, srcIp=212.129.27, srcPort=43324, dstIp=59.67.152, dstPort=47137, attackInfo=SYN-Flood, flag=SYN_INVALID]
[date=2015-12-24 07:22:15, srcIp=212.129.27, srcPort=27950, dstIp=59.67.152, dstPort=27328, attackInfo=SYN-Flood, flag=SYN_INVALID]
[date=2015-12-24 07:22:45, srcIp=212.129.27, srcPort=27950, dstIp=59.67.152, dstPort=27328, attackInfo=SYN-Flood, flag=SYN_INVALID]
[date=2015-12-24 07:28:50, srcIp=212.129.27, srcPort=39676, dstIp=59.67.152, dstPort=9521, attackInfo=SYN-Flood, flag=SYN_INVALID]
[date=2015-12-24 07:29:21, srcIp=212.129.27, srcPort=39676, dstIp=59.67.152, dstPort=9521, attackInfo=SYN-Flood, flag=SYN_INVALID]
[date=2015-12-24 07:29:21, srcIp=212.129.27, srcPort=27950, dstIp=59.67.152, dstPort=27819, attackInfo=SYN-Flood, flag=SYN_INVALID]
[date=2015-12-24 07:29:51, srcIp=212.129.27, srcPort=27950, dstIp=59.67.152, dstPort=27819, attackInfo=SYN-Flood, flag=SYN_INVALID]
[date=2015-12-24 07:30:21, srcIp=212.129.27, srcPort=27950, dstIp=59.67.152, dstPort=27819, attackInfo=SYN-Flood, flag=SYN_INVALID]
[date=2015-12-24 07:33:55, srcIp=212.129.27, srcPort=39676, dstIp=59.67.152, dstPort=9562, attackInfo=SYN-Flood, flag=SYN_INVALID]
[date=2015-12-24 07:34:25, srcIp=212.129.27, srcPort=39676, dstIp=59.67.152, dstPort=9562, attackInfo=SYN-Flood, flag=SYN_INVALID]
```

Fig. 3. Duration of up to 9 hours 27 minutes of SYN-flood attacks (the part of it) .

The attacker (211.129.27.x) repeatedly transform the source port number to port number, while the host 59.67.152.x issued the total 493 illegal SYN flow. This persistent SYN attack can bypass the DDoS detection method based on the number of flow SYN real-time statistics. Only through the retrospective analysis of historical data, it can accurately locate the attack events. From December 24th 2015 to December 29th 2015, persistent SYN attack appeared in a total of 8 times, accounted for 9.1% of the total number of SYN Flood attack. The attack detection and prevention must more attention to this type SYN attacks.

Fig. 4. shows the changes in the record_creation_num and ack_num (the flow number of flag ACK per unit of time) during the 5:04 to 7:08. By comparison, it can be seen that in this period. ACK netflow dominates the trend of the overall network load. In the process of data stream processing, the router will record all TCP flag turn to a Netflow records, so a legitimate TCP connection and its corresponding Netflow records should at least contain both max and min marks. The typical feature of ACK-Flood is that the attacker will send a large number of IP packets containing only flag max in a short time. The victim host of the receiver to check if the data packets turn to the effectiveness of TCP flag, when the data input speed much faster than the victim host processing speed, the victim system will collapse. When the network traffic data appeared in a large number of ACK flow, it is likely to herald the ACK attack against the current computer network.

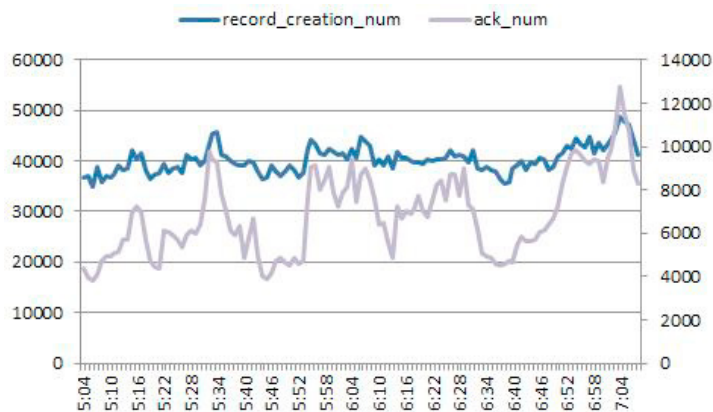


Fig. 4. the change trend of record_creation_num and ack_num.

Fig. 5. shows the source host in unit time issued a number of ACK flow distribution ratio(ack_src_less_5、ack_src_less_10、ack_src_less_20 and ack_src_large_20) respectively represent unit time issued ACK flow number less than 5, between 5 and 10, between 10 and 20 and greater than 20 hosts a number. It can be found that ack_src_less_10 has a very significant change in the ACK-Flood attack.

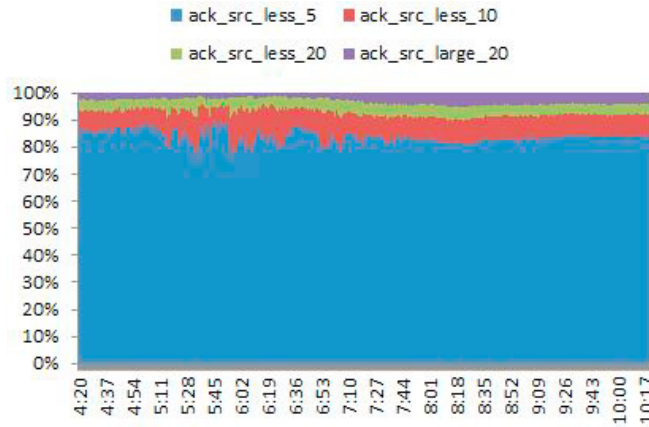


Fig. 5. proportional distribution of flow ACK number in the same IP unit time.

Fig. 5. shows the Icmp_num analysis data on December 25th 2015 to December 28th 2015, which is different with the SYN Flood attack. Tianjin Education metropolitan area network of ICMP-Flood attacks focused in the morning 11:00 to 18:00 pm. In this period of time, it shows that in the morning the data relatively stable. When the time of 12:00 to 18:00 exactly is a high incidence of NTP reflection attacks, which from the side proved that the attacker will usually combined NTP flood attacks with the ICMP flood attacks.

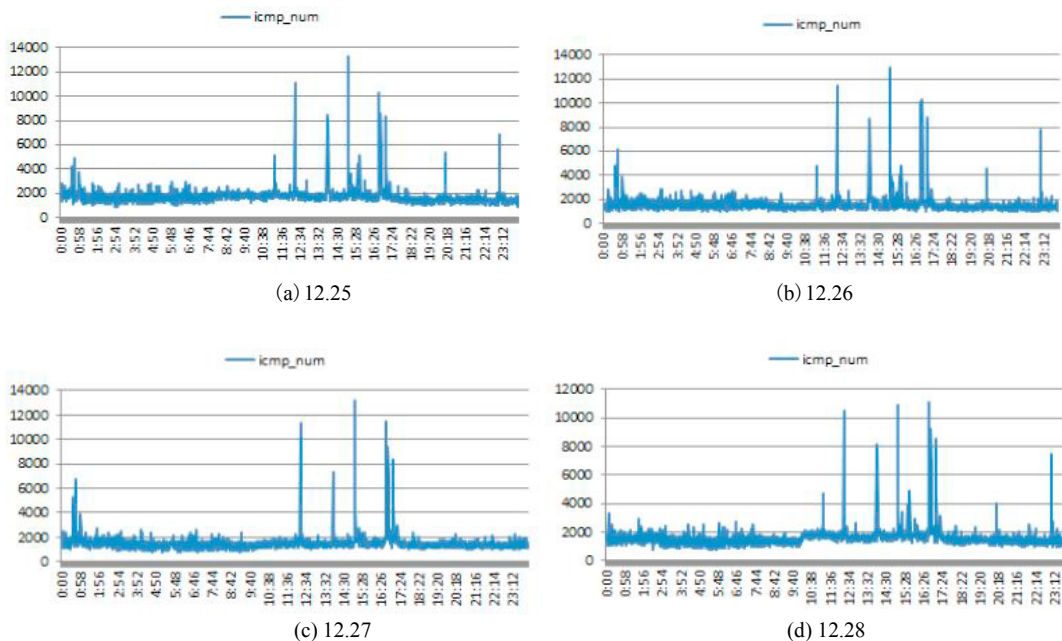


Fig. 6. the Icmp_num data of December 25th 2015 to December 28th 2015.

Fig. 6. shows the Icmp_num analysis data on December 25th 2015 to December 28th 2015, which is different with the SYN Flood attack. Tianjin Education metropolitan area network of ICMP-Flood attacks focused in the morning 11:00 to 18:00 pm. In this period of time, it shows that in the morning the data relatively stable. When the time of 12:00 to 18:00 exactly is a high incidence of NTP reflection attacks, which from the side proved that the attacker will usually combined NTP flood attacks with the ICMP flood attacks.

4. Conclusions

In this paper, an anomaly traffic detection system for DDoS attack is proposed, which is composed of a number of weighted moving average (EWMA) algorithm, an anomaly traffic interval location algorithm. EWMA algorithm core idea is to use previous moment estimation value and the current observation value to calculate the estimated value. Through actual measurement network comparison of load index values and the corresponding EWMA estimation value discrimination network traffic, we can achieve the results whether there are abnormal changes. According to the parameters, the anomaly flow interval location algorithm can accurately identify the occurrence period of anomaly traffic such as the flow direction and the event priority.

In this paper, the performance and effectiveness of the traffic real-time monitoring system are verified. First of all, through a set of comparative experiments, this paper analyzes and discusses the number of system nodes and so on. Secondly, this paper combines the network traffic data of Tianjin education metropolitan area network with the NSFOCUS ANTI-DDoS system, data processing, data analysis and other steps of operation to complete the detection rate of some type attacks. Finally, this paper analyzes the network traffic of TERNET from record_creation_num, syn_to_flow_ratio, ack_num, icmp_num and other network load indicators.

Distributed middleware based network security analysis technology is still in its infancy, there is no mature and perfect theory system. Combining data technology and network security analysis, system implementation and other aspects made some attempts. Though it has initially solved the system performance bottleneck, a variety of computing mode hybrid uses common problems. Further research that dynamic interactive visualization is needed. The system can be further studied and the performance can be better optimized.

References

- [1] Patcha A, Park J, An overview of anomaly detection techniques: Existing solutions and latest technological trends, *Computer Networks*, 2007, 51(12) : 3448-3470
- [2] Laskov P, Gehl C, Kruger S, Multer K, Incremental support vector learning: Analysis, implementation and applications, *Journal of Machine Learning Research*, 2006, 7 : 1909-1936 .
- [3] Thomas D, Arno W, Bernhard P, A Framework for Real-Time Worm Attack Detection and Backbone Monitoring [J]. *Proceedings of the 2005 First IEEE International Workshop on Critical Infrastructure Protection*. 2005
- [4] Tran Q, Jiang F, Hu J. A Real-Time NetFlow-based Intrusion Detection System with Improved BBNN and High-Frequency Field Programmable Gate Arrays[C]// *IEEE 11th International Trust, Security and Privacy in Computing and Communications (TrustCom)*. Liverpool, England. 2012:201-208.
- [5] Wei G, Research and development of distributed stream data real-time computing framework [D], *Zhejiang University of Technology*, 2013
- [6] Gang Z, Research on network security intrusion scenario reconstruction technology based on cloud computing platform[D], *Nanjing University of Technology*, 2013
- [7] Yan H, Distributed intrusion detection system for large scale network [D]. *University of the Chinese Academy of Sciences*, 2003.
- [8] Miao L, Ding W, Gong J. A real-time method for detecting internet-wide SYN flooding attacks[C]//*IEEE International Workshop on Local and Metropolitan Area Networks (LANMAN)*. Beijing, China, 2015: 1-6.
- [9] Wang J. Properties of EWMA Controllers with Gain Adaptation [J]. *IEEE Transactions on Semiconductor Manufacturing*, 23(2): 159-167.
- [10] Zhang L, Chen G, Castagliola P, On t and EWMA t charts for monitoring changes in the process mean. *Quality and Reliability Engineering International*, 2009, 25, (8): 933-945.
- [11] Kazemzadeh R. B, Karbakhani M , An EWMA t chart with variable sampling intervals for monitoring the process mean. *The International Journal of advanced Manufacturing Technology*, 2013 , 66 (1-4): 125-139.
- [12] Wang H, Sun H, Li. C, Rahnamayan S, Diversity enhanced particle swarm optimization with neighborhood search *Information Sciences*, 2013, 223: 119-135.
- [13] Idhammad M, Afdel K, Belouch M (2017) Dos detection method based on artificial neural networks. *Int J Adv Comput Sci Appl (ijacasa)* 8(4):465-471
- [14] Bhuyan MH, Bhattacharyya DK, Kalita JK (2015) An empirical evaluation of information metrics for low-rate and high-rate ddos attack detection. *Pattern Recogn Lett* 51:1-7
- [15] Yue X, Wang J, Huang W (2020). Hybrid fuzzy integrated convolutional neural network (HFICNN) for similarity feature recognition problem in abnormal netflow detection. *Neurocomputing*, 415, 332 – 346.
- [16] Elejla OE, Anbar M, Belaton B (2017). ICMPv6-Based DoS and DDoS Attacks and Defense Mechanisms: Review. *IETE Technical Review (Institution of Electronics and Telecommunication Engineers, India)*, 34(4), 390 – 407.